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Fm16

Int 9

Review

1 $\int (e^x - \frac{1}{e^x})^2 \, dx = \int e^{2x} - 2 + e^{-2x} \, dx$

$= \frac{1}{2}e^{2x} - 2x - \frac{1}{2}e^{-2x}$

2 $a = 5$

$\frac{5}{1-r} = 6.25$

$5 = 6.25 - 6.25r$

$\frac{-1.25}{-6.25} = r$

$r = \frac{1}{5}$

3 $5^x = e^{\ln 5 \, x}$

$\therefore \frac{d}{dx} 5^x = \ln 5 \, e^{\ln 5 \, x}$

$= 5^x \ln 5$

4 $\int_0^1 \frac{1 - e^{-3x}}{1 + e^{-3x}} dx = \int \frac{1}{1 + e^{-3x}} \, dx - \int \frac{e^{-3x}}{1 + e^{-3x}} \, dx$

$= -\frac{1}{3}\int \frac{1}{1+e^u} \, du \quad + \frac{1}{3}\ln|1 + e^{-3x}|$

$-\frac{1}{3}\int 1 - \frac{e^u}{1+e^u} \, du$

$= -\frac{1}{3}u + \frac{1}{3}\ln|1 + e^u| \quad + \frac{1}{3}\ln|1 + e^{-3x}|$

$= -\frac{1}{3} \cdot -3x + \frac{1}{3}\ln|1 + e^{-3x}| + \frac{1}{3}\ln|1 + e^{-3x}|$

$= x + \frac{2}{3}\ln|1 + e^{-3x}|$

$\therefore \left[x + \frac{2}{3}\ln|1 + e^{-3x}|\right]_0^1$

$= 1 + \frac{2}{3}\ln|1 + e^{-3}| - \left(0 + \frac{2}{3}\ln|2|\right)$

$= 1 + \frac{2}{3}\ln\left(\frac{e^{-3}+1}{2}\right)$

5 $y = 3\sin\theta \quad x = 5\cos\theta$

$y^2 = 9\sin^2\theta \quad x^2 = 25\cos^2\theta$

$= 25(1 - \sin^2\theta)$

$= 25 - 25\sin^2\theta$

$\frac{25 - x^2}{25} = \sin^2\theta$

$y^2 = 9\left(\frac{25 - x^2}{25}\right)$

$\frac{y^2}{9} = \frac{25 - x^2}{25}$

$= 1 - \frac{x^2}{25}$

$\frac{x^2}{25} + \frac{y^2}{9} = 1$

Consolidation

1 $\frac{1}{y-1}\,dy = x\,dx$

$\ln|y-1| = \frac{1}{2}x^2 + c$

$y-1 = e^{\frac{1}{2}x^2+c}$

$y = e^{\frac{1}{2}x^2+c} + 1$

2 $(x-1)\frac{dy}{dx} = xy$

$\frac{x}{x-1}\,dx = \frac{1}{y}\,dy$

$\frac{x}{x-1} = \frac{x-1+1}{x-1}$

$= 1 + \frac{1}{x-1}$

$x + \ln|x-1| = \ln|y| + c$

@ (3, 1):

$3 + \ln 2 = 0 + c$

$c = 3 + \ln 2$

$\therefore x + \ln|x-1| = \ln|2y| + 3$

3 $(1-e^{2y})\frac{dy}{dx} = e^{y}$

$(e^{-y} - e^{y})\,dy = 1\,dx$

$-e^{-y} - e^{y} = x + c$

@(2, 0):

$-e^{0} - e^{0} = 2 + c$

$-2 = 2 + c$

$c = -4$

$-e^{-y} - e^{y} = x - 4$

4 $x^2\frac{dy}{dx} = y^2$

$\frac{x^2}{dx} = \frac{y^2}{dy}$

$x^{-2}\,dx = y^{-2}\,dy$

$-\frac{1-cx}{x} = -\frac{1}{y}$

$-\frac{x}{1-cx} = -y$

$y = \frac{x}{1-cx}$

5 $\frac{dC}{dt} = kC$ ① @ $t=0$, $C=100$

@ $t=2$, $C=500$

$\frac{1}{C}\,dC = k\,dt$

$\ln|C| = kt + c$

$C = e^{kt}e^{c}$

$e^{c} = 100$ (from ①)

$C = 100e^{kt}$

$500 = 100e^{2k}$

$e^{2k} = 5$

$e^{k} = \sqrt{5}$

$\rightarrow C = 100 \times 5^{\frac{1}{2}t}$

6 at $t=5$:

$C=100\times 5^{\frac{1}{2}\times 5}$

$=100\times 5^{2.5}=5590$

c) $100\times 5^{\frac{1}{2}t}>5000$

$5^{\frac{1}{2}t}>50$

$\frac{1}{2}t>\log_5 50$

$t>\log_5 2500$

>4.86

d $100\times 5^{\frac{1}{2}t}>150$

$5^{\frac{1}{2}t}>1.5$

$\frac{1}{2}t>\log_5 1.5$

>0.252

$t>0.504$

6 a) $x^{\frac{3}{2}}dx=-\frac{50}{\pi}dt$

$\frac{2}{5}x^{\frac{5}{2}}=-\frac{50}{\pi}t+c$

@ $t=0$, $x=25$

$\therefore c=\frac{2}{5}\times 25^{\frac{5}{2}}$

$=1250$

$\frac{2}{5}x^{\frac{5}{2}}=-\frac{50}{\pi}t+1250$

b) @ $x=0$:

$\frac{50}{\pi}t=1250$

$t=25\pi$

$=78.5s$

7 a) $\frac{dT}{dt}=-k(T-T_0)$ where T_0 = ambient room temp

$T_0=20$

$\frac{dT}{dt}=-k(T-20)$

$\frac{1}{T-20}dT=-k\,dt$

$\ln|T-20|=c-kt$

$T-20=e^{c-kt}$

$T=e^{c-kt}+20$

@ $t=0$, $T=84$

$e^c=64$

$c=\ln 64$

$\therefore T=e^{6\ln 2-kt}+20$

@ $t=300$, $T=52$

$52=e^{6\ln 2-300k}+20$

$e^{6\ln 2-300k}=32$

$6\ln 2-300k=5\ln 2$

$300k=\ln 2$

$k=\frac{\ln 2}{300}$

$\therefore T=e^{6\ln 2-\frac{\ln 2}{300}t}+20$

$21=e^{6\ln 2-\frac{\ln 2}{300}t}+20$

$6\ln 2-\frac{\ln 2}{300}t=0$

$\frac{\ln 2}{300}t=6\ln 2$

$t=1800s$

$=30$ mins

b) Model never actually reaches 20° (ambient room temperature) but the tea will.

Exam Qs

1 @$t=0$, $N=20$

$$\frac{dN}{dt} = \frac{1}{1000}(10000 - N^2)$$

$$\frac{1}{10000 - N^2}\, dN = \frac{1}{1000}\, dt$$

$$\hookrightarrow \frac{A}{(100+N)} + \frac{B}{(100-N)} = \frac{1}{10000-N^2}$$

$A(100-N) + B(100+N) = 1$

$100A - AN + 100B + BN = 1$ $\qquad$ $BN - AN = 0$

$A = B$

$\therefore 200A = 1$

$A = \frac{1}{200} = B$

$$\therefore \frac{1}{200}\ln|100+N| - \frac{1}{200}\ln|100-N| = \frac{1}{1000}t + c$$

$$\ln|100+N| - \ln|100-N| = \frac{1}{5}t + c$$

$$\frac{100+N}{100-N} = e^{0.2t+c}$$

$$\frac{120}{80} = e^c$$

$$\frac{3}{2} = e^c$$

$$\frac{100+N}{100-N} = \frac{3}{2}e^{0.2t}$$

$$100 + N = 150e^{0.2t} - \frac{3}{2}Ne^{0.2t}$$

$$N\left(1 + \frac{3}{2}e^{0.2t}\right) = 150e^{0.2t} - 100$$

$$N = \frac{150e^{0.2t} - 100}{1 + \frac{3}{2}e^{0.2t}}$$

$$= \frac{300e^{0.2t} - 200}{3e^{0.2t} + 2}$$

$$= \frac{100(3e^{0.2t} - 2)}{3e^{0.2t} + 2}$$

b As t increases, N will approach 100

c May not account for predators, famine, abundance of food, etc.

2 $\frac{dy}{dx} = \frac{x^2 \sin 2x}{2\cos^2 4y - 1}$

$2\cos^2 4y - 1 = \cos 8y$

$\therefore \cos 8y \, dy = x^2 \sin 2x \, dx$

$\hookrightarrow \int x^2 \sin 2x \, dx$

$= -\frac{1}{2}x^2 \cos 2x + \int x \cos 2x$

$= -\frac{1}{2}x^2 \cos 2x + \left(\frac{1}{2}x \sin 2x - \frac{1}{2}\int \sin 2x \, dx\right)$

$+ \frac{1}{2}x \sin 2x + \frac{1}{4}\cos 2x$

$\frac{1}{8}\sin 8y = \frac{1}{2}x \sin 2x + \frac{1}{4}\cos 2x - \frac{1}{2}x^2 \cos 2x + C$

$\sin 8y = 4x \sin 2x + 2\cos 2x - 4x^2 \cos 2x + A$

3 @ $t=0$: $V = 800$

$\frac{dV}{dt} = -20$

$-20 = 500k \quad k = \frac{2}{50} = -\frac{1}{25}$

$\frac{dV}{dt} = kV$ $\quad$ @ $t = 0$, $k = \frac{1}{25} = 0.04$

$\ln|V| = kt + c$

$V = Ae^{kt} \quad (A = e^c)$

$500 = Ae^0 = A$

$e^c = 500$

$\therefore V = 500e^{kt}$

$= 500e^{-\frac{1}{25}t}$

$250 = 800e^{-\frac{1}{25}t}$

$\frac{1}{2} = e^{-\frac{1}{25}t}$

$\ln 2 = \frac{1}{25}t$

$t = 25\ln 2$

$= 17.3$ hours

b) the ambient temp. outside snowball does not change.